

Temba-Panoptic

Video Panoptic Segmentation via Vision Transformers with Temporal-Mamba modules

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1. Overview

One of the main challenges present in **Video Panoptic Segmentation** is dealing with long-range **time dependencies** and considering temporal consistency. To address this problem, we propose **Temba-Panoptic**, a hybrid architecture that combines a strong panoptic segmentation backbone (Mask2Former) with Temba blocks used as a lightweight temporal query-memory module.

2. Dataset

The dataset we choose is **JRDB-PanoTrack**; it contains 4000 360-degree panoramic frames generated by merging five camera views, creating 54 videos with 2D segmentation masks and temporal labeling.

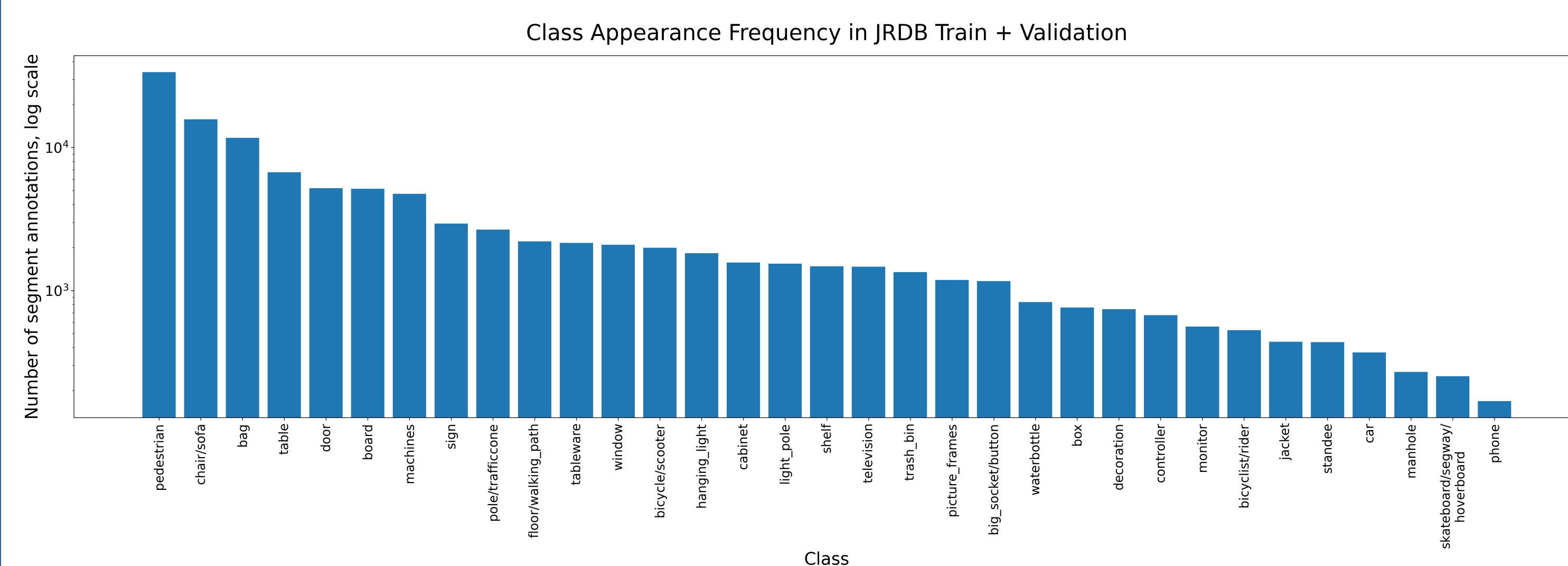


Figure 1: Frequency (in log scale) of each class in the dataset.

3. Temba block architecture

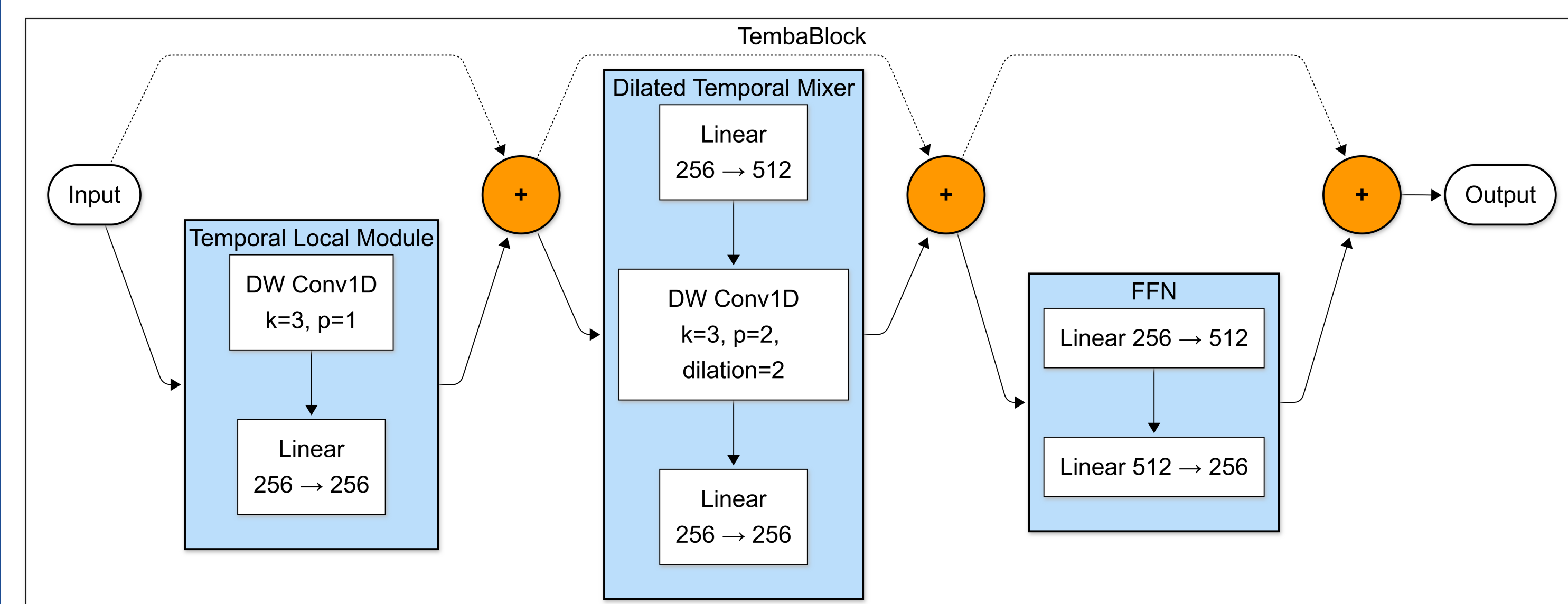


Figure 2: Basic scheme of Temba blocks using convolutional layers to extract temporal information.

4a. Methodology: training setup

Loss function:

$$\mathcal{L} = 2 \cdot l_{CE} + 5 \cdot l_{MASK} + 5 \cdot l_{DICE}.$$

Two approaches:

- Add **more transformer blocks** to M2F, from 9 to 12;
- Add the **Temba adapters** and test what happens when increasing the number of Temba blocks in each adapter, with $N = 1, 2, 3$.

4b. Methodology: evaluation

Evaluation metrics:

1. **Panoptic Quality** (PQ);
2. **Segmentation and Tracking Quality** (STQ);
3. **Video Panoptic Quality** (VPQ);
4. Number of **tracked objects** in the video.

5. Results

Adding **one Temba block** brings improvements across nearly all metrics; however, adding more blocks increases the number of **convolutional operations** that over-smooth the features arriving in the decoder, decreasing the metrics.

Table 1: Main results on JRDB-PanoTrack dataset.

Method	PQ	PQ _{SQ}	PQ _{RQ}	STQ	STQ _{SQ}	STQ _{AQ}
Baseline	3.57	14.89	4.82	17.74	5.99	52.54
+ 3 dec. lay.	3.00	14.92	4.05	17.81	6.10	52.79
TEMBA M2F						
$N = 1$	4.58	19.94	6.40	19.93	7.63	52.08
$N = 2$	3.46	21.47	4.58	17.15	5.62	52.36
$N = 3$	3.69	19.85	5.06	19.31	7.09	52.60

Method	VPQ	VPQ-1	VPQ-2	VPQ-4	VPQ-8	#tracks
Baseline	3.46	4.47	3.54	3.05	2.80	85
+ 3 dec. lay.	3.33	4.27	3.47	2.94	2.66	96
TEMBA M2F						
$N = 1$	3.90	4.93	4.04	3.55	3.10	87
$N = 2$	3.21	4.23	3.34	2.78	2.50	94
$N = 3$	3.18	3.81	3.13	2.93	2.84	90

Improvements of 1 Temba block model with respect to baseline

PQ	PQ _{SQ}	PQ _{RQ}	STQ	STQ _{SQ}	STQ _{AQ}	VPQ	VPQ-1	VPQ-2	VPQ-4	VPQ-8	#tracks
$\times 1.28$	$\times 1.34$	$\times 1.33$	$\times 1.23$	$\times 1.27$	$\times 0.99$	$\times 1.27$	$\times 1.10$	$\times 1.14$	$\times 1.16$	$\times 1.11$	+2

Varying the number of iterations, we see that adding more iterations initially **increases** all the metrics, then all the values start to **decrease**, while at the end all metrics **increase again**. Hence, increasing the number of convolutional operations over-smooths the features, causing a decrease in the metrics. However, adding many training iterations allows the Temba blocks to extract more temporal information, explaining the final rise in the metrics.

Table 2: Evolution of validation metrics across training iterations for the 2-Temba blocks model.

# iter.	PQ	PQ _{SQ}	PQ _{RQ}	STQ	STQ _{SQ}	STQ _{AQ}	VPQ	VPQ-1	VPQ-2	VPQ-4	VPQ-8	#tracks
6k	4.56	19.00	4.65	18.10	6.22	52.66	3.08	4.00	3.21	2.66	2.45	84
12k	3.42	21.04	4.52	18.17	6.25	52.53	3.97	3.13	3.97	3.20	2.61	88
18k	3.18	19.64	4.24	17.03	5.48	52.93	3.10	3.74	3.12	2.88	2.69	87
24k	3.13	18.41	4.21	17.15	5.58	52.71	2.86	3.61	3.04	2.50	2.29	93
30k	3.46	21.47	4.58	17.15	5.62	52.36	3.21	4.23	3.34	2.78	2.50	94

Temba-Mask2Former architecture

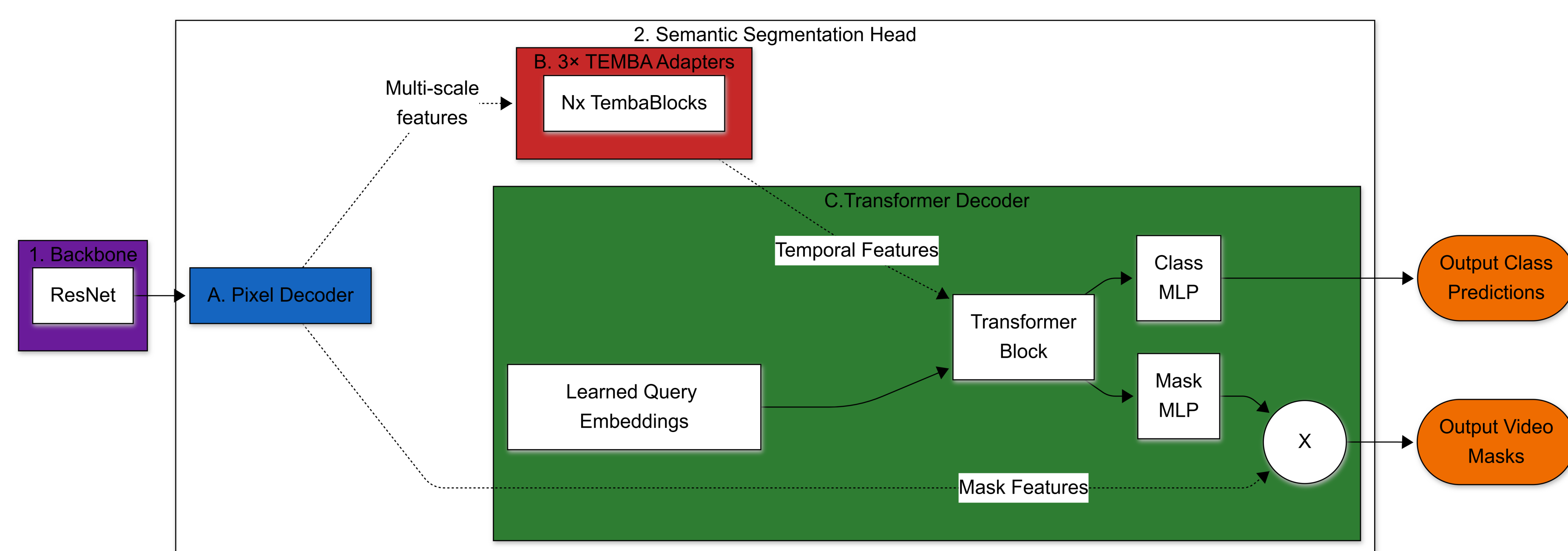


Figure 3: Temba-Mask2Former architecture enriching features of the pixel decoder with temporal scale through Temba Blocks.